

CSE 483 Gezi Final

Olcay Taner YILDIZ

Write an application that allows the user to update info about a chapulcu. The info consists of the city that the chapulcu is from, the fragment of city that the chapulcu chapuls, the number of gas capsules (G) she can inhale before she collapses, the number of gas capsules she has taken (T), and the frustration factor (F). Initially chapulcu is not chapulling therefore the number of gas capsules she has taken is ($T = 0$), but she is frustrated about the politicians and therefore $F = 0.5$. The health and happiness factors of the chapulcu is also displayed, but can not be edited and is dependent on (G , T , and F) as

$$\begin{aligned} Health &= 100 \times (1.2 - \frac{T}{G}) \\ Happiness &= Health - F \times 20 \end{aligned} \quad (1)$$

- 1) (16 points) Design the interface. Make the city and city fragment fields editable. The number of capsules she can inhale (G) (initial value random between 1 and 10), the number of gas capsules she has taken (T) (initial value 0) are not editable.
- 2) (8 points) Add Java codes to save city and city fragment states (Use Bundle).
- 3) (6 points) Add a button to chapul. When the user presses the button, the chapulcu chapuls and receives gas capsule from police. Corresponding new health value and happiness factor will be displayed.
- 4) (6 points) Add a button that will display info from a web browser about the city the chapulcu is from. Programmatically open google with the city info being searched.
- 5) (6 points) Use Alert Dialog to warn the user if she wants to chapul although she has collapsed (in case of $G = T$).
- 6) (8 points) Use SharedPreferences to save the info about the numbers G and T .
- 7) (12 points) Add one Imageview to the application which displays a picture about the city fragment the chapulcu is from. When the name is changed the program will look the assets folder and if the picture exists it will display the picture. The pictures of the cities are stored in the assets folder as Taksim.jpg, Besiktas.jpg etc.
- 8) (12 points) Prepare an options menu. Add a menu item to change the number of capsules the chapulcu can inhale (G). When the user selects the menu item, create a custom GUI for a new Alert Dialog to allow the user to enter G .
- 9) (6 points) Add an alert sound when happiness is reduced to 20 or less.
- 10) (8 points) When the user shakes the phone, the chapulcu gets help from other chapulcus that is the number of gas capsules she has taken decreases by 1.
- 11) (12 points) Add another option menu item to save chapulcu info into the database of Republic of Toma Entertainers. All info about the current chapulcu are stored in the database.